

17. (3 points)

Given the following financial information for a property and casualty insurance company:

- Projected ROE = 9.0%.
- Projected dividend payout rate = 40.0%.
- Yield on long-term T-bonds = 4.7%.
- Term premium for long-term T-bonds = 1.3%.
- Expected equity risk premium = 7.0%.
- Insurer's equity beta = 0.8.

a. (1 point)

Calculate the company's Price to Earnings (P-E) Ratio assuming projections continue in perpetuity.

b. (0.5 point)

Explain the effect on the company's P-E ratio assuming the dividend payout rate changes to 60%.

c. (0.5 point)

Explain the effect on the company's P-E ratio assuming the yield on long-term T-bonds is 4% and the equity risk premium is unchanged.

d. (1 point)

Describe two alternative uses for the P-E ratio.

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SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

QUESTION 17	
TOTAL POINT VALUE: 3	LEARNING OBJECTIVE(S): B2, B3
SAMPLE ANSWERS	
Part a: 1 point	
<u>Sample 1</u> $P/E = 1 - \rho / (k - \rho * ROE)$ $ROE = 0.09$ $r_f = 0.047 - 0.013 = 0.034$ $k = 0.034 + 0.8 (0.07) = 0.09$ $\rho = 1 - 0.4 = 0.6$ $P/E = (1 - 0.6) / (0.09 - 0.6(0.09)) = 11.11$ <u>Sample 2</u> $P-E = Div / (k - g)$ $k = 0.047 - 0.013 + 0.8(0.07) = 0.09$ $g = 0.09(1 - 0.4) = 0.054$ $P-E = 0.4 / (0.09 - 0.054) = 11.11$	
Part b: 0.5 point	
<u>Sample 1</u> Div rate $\uparrow$ 60%, plowback rate $\downarrow$ 40%, $g = 0.09 * 0.4 = 3.6\%$ $P/E = 0.6 / (0.09 - 0.036) = 11.111$ Assuming that what's not paid in dividend is reinvested in the firm, the P/E ratio stays constant as increase in dividend rate is offset by a decrease in growth rate. <u>Sample 2</u> Since the discount rate equals the ROE (both 9%), the dividend payout rate has no effect on the P/E ratio. It will remain 11.11. <u>Sample 3</u> $P/E = 0.6 / (0.09 - 0.036) = 11.11 \rightarrow$ No Effect $G = 0.09(0.4) = 0.036$	
Part c: 0.5 point	
<u>Sample 1</u> $r_f = 0.04 - 0.013 = 0.027$ $K = 0.027 + 0.8(0.07) = 0.083$ $P/E = 0.4 / (0.083 - 0.054) = 13.79$ When r_f decreases $\Rightarrow$ cost of capital decrease $\Rightarrow$ PV of dividends increase providing higher multiple	

SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

Sample 2

This would decrease r_f , which would decrease k , ultimately decreasing the denominator of the ratio -> Increases P/E ratio.

Sample 3

$$r = 0.04 - 0.013 = 0.027$$

$$k = 0.027 + 0.8(0.07) = 0.083$$

$$P/E = 0.4 / (0.083 - 0.054) = 13.793$$

Increases it

Part d: 1 point

Sample 1

- Validation of assumptions. You can compare your P/E ratio to that of similar firms. If they are reasonably close, then you will feel more confident in your assumptions. If they are not close, then this may indicate a need to revisit assumptions
- The P/E ratio can assist in the calculation of the terminal value. The terminal value has a large influence in the overall value of the company & is highly leveraged. You could use the P/E ratio to remove some of the uncertainty in this calculation

Sample 2

- Shortcut to valuation -> allows us to quickly compute company value without needing detailed forecast of financial projections (time consuming & difficult to project NI over multiple years)
- Diagnostic tool for valuation -> Can compare to peer companies to assess reasonability. Also, can compare to result from DDM AE, FCFE method to compare/assess reasonability of terminal value assumptions

Sample 3

- You can use P-E ratios to check assumptions used in other valuation models. If the results of the valuation models don't produce P-E ratios in line with peer companies, you may want to revisit model assumptions
- You can price/value your company quickly using peer/competitor multiples of P-E ratios. Finding comparable peers by looking at premium volume, type of business and expected return will help you select which peer P-E ratios are appropriate

Sample 4

- The P-E ratio can be used as a diagnostic check on an alternative valuation, such as the dividend discount model.
- The P-E ratio could be used for the terminal value calculation in conjunction with another valuation method. For example, it could be used with the FCFE method, to calculate the terminal value

Sample 5

- Can use the P-E ratio to compare valuation with similar firms and validate assumptions
- Can use the P-E ratio to guide the calculation of the terminal value in other valuation methods

SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

EXAMINER'S REPORT
The candidate was expected to demonstrate knowledge of firm valuation and how the P/E ratio can play an integral role in this exercise
Part a
The candidate was expected to determine the discount rate using the Capital Asset Pricing Model and apply the Dividend Discount Model to calculate the company's Price to Earnings Ratio
Common mistakes included: • Neglecting to subtract the term premium when calculating the risk-free rate • Subtracting the term premium from the equity risk premium in the CAPM formula • Failing to recognize the application of the Dividend Discount Model to calculate the P/E ratio (using other valuation models) • Using incorrect formula for Dividend Discount Model
Part b
The candidate was expected to recognize that the dividend payout ratio has no impact on the P/E ratio when the discount rate equals the company's return on equity
Common mistakes included: • Failing to adjust the growth rate for the increased dividends • Only recognizing the impact on growth, but not reflecting the increased dividends • Failing to recognize the significance of the discount rate being equal to the ROE
Part c
The candidate was expected to recognize that the P/E ratio increases when the discount rate decreases
Common mistakes included: • Adjusting the ROE as well as the discount rate • Failing to recognize the lower discount rate would increase the PV of the future dividends & earnings
Part d
The candidate was expected to list and describe two alternative uses of the P/E ratio
Candidates did not receive credit for use cases that did not relate to valuation, as the P/E ratio is a relative measure of company value to net income
Common mistakes included: • Failing to recognize the use of P/E as a valuation metric • Failing to elaborate on the uses provided • Misinterpreting question and providing other valuation ratios (e.g. P/BV) • Vagueness in answers (e.g. "benchmarking")